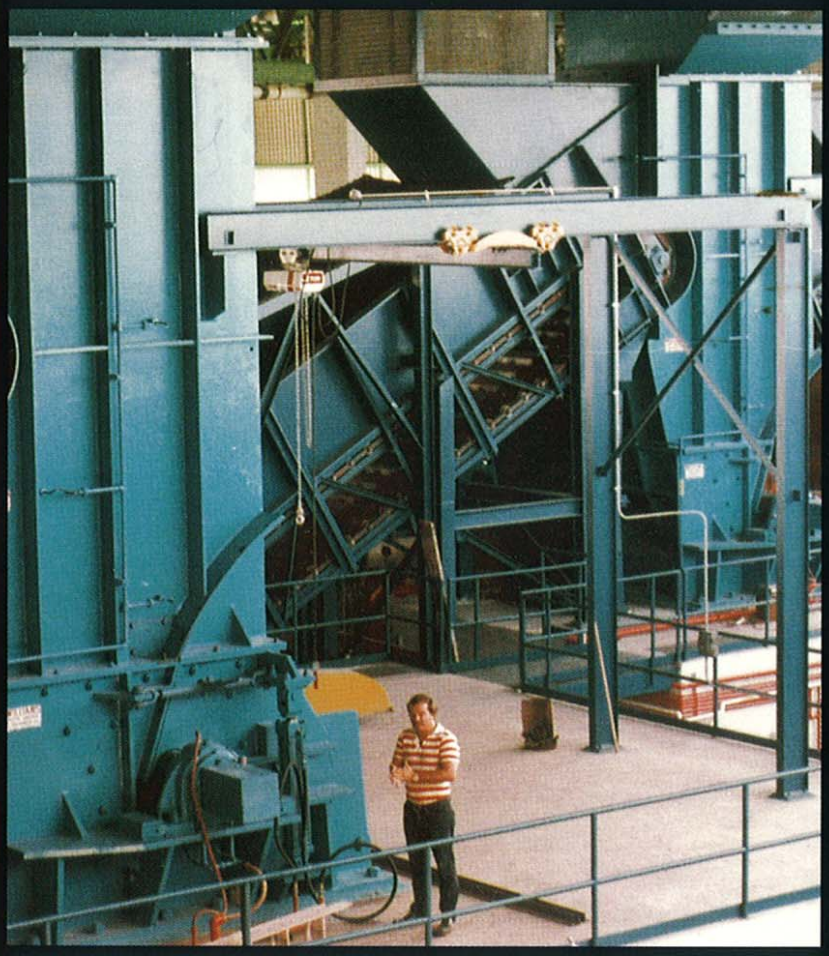
"NO-NIFE" HOGS



WILLIAMS

Patent Crusher & Pulverizer Company
2701 North Broadway, St. Louis, MO U.S.A. 63102
Area Code 314/621-3348 FAX 314/436-2639
E-Mail: www.crusher@inlink.com

WILLIAMS "NO-NIFE" HOGS & SHREDDERS

...Shred Bark & Wood Waste By Impact
...Eliminate Shutdowns To Sharpen Knives
...Eliminate Costly Tool Room Labor

Williams "No-Nife" Hogs are proven performers in the forest products industry. Economical size reduction in the "No-Nife" Hog is the first important step for bark and wood waste whether the product is to be sold, manufactured into secondary products, or used as fuel in modern smokeless boilers.

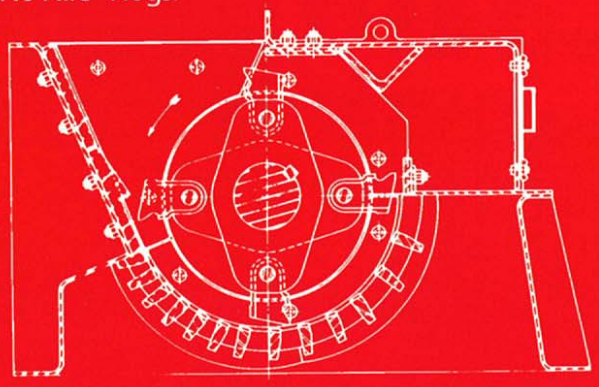
The hogged product can be sold for garden mulch, animal bedding, and litter. Or it can be used in the manufacture of particle board and fireplace logs.

In the pulp industry, Williams "No-Nife" Hogs are extensively used to reduce bark and broken pulpwood following drum, hydraulic, steam, or mechanical debarkers.

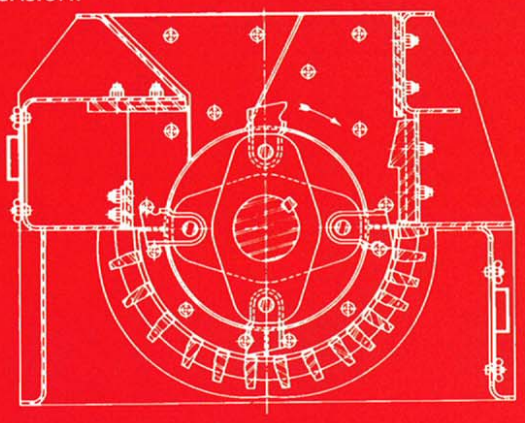
Money saving automation is possible in the conveying and handling of hogged material due to its uniform size and superior storage properties.

Williams "No-Nife" Hogs are well known for their reliability, rugged construction, and long operating campaigns. And knife sharpening has been eliminated because there are NO KNIVES in a Williams "NO-NIFE" Hog.

Section is typical of the construction furnished in a "No-Nife" Hog for primary reduction of wood waste. Swing hammers are free to pivot when uncrushables are encountered. Metal trap is located for convenient access. Large unrestricted feed opening is common to all "No-Nife" Hogs.



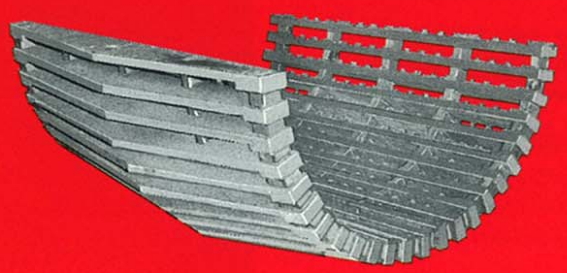
Typical of the construction furnished in an over-running "No-Nife" Hog. Swing hammers are furnished as standard. A full 180 degree cage permits unrestricted discharge and assures high output for a given rotor dimension.



ADVANCED CAGE BAR DESIGN

Williams Hogs are equipped with advanced design cage bars. Where product requirements change, the cage can be field modified without the need for special tools. Cast grid sections can be supplied, or where fine product is desired, perforated metal cages are available.

Shown below at the left is a typical cage assembled from single bevel edged bars. This design, one of the most frequently furnished, provides an efficient shearing action which contributes to high output and low power per ton of product handled.

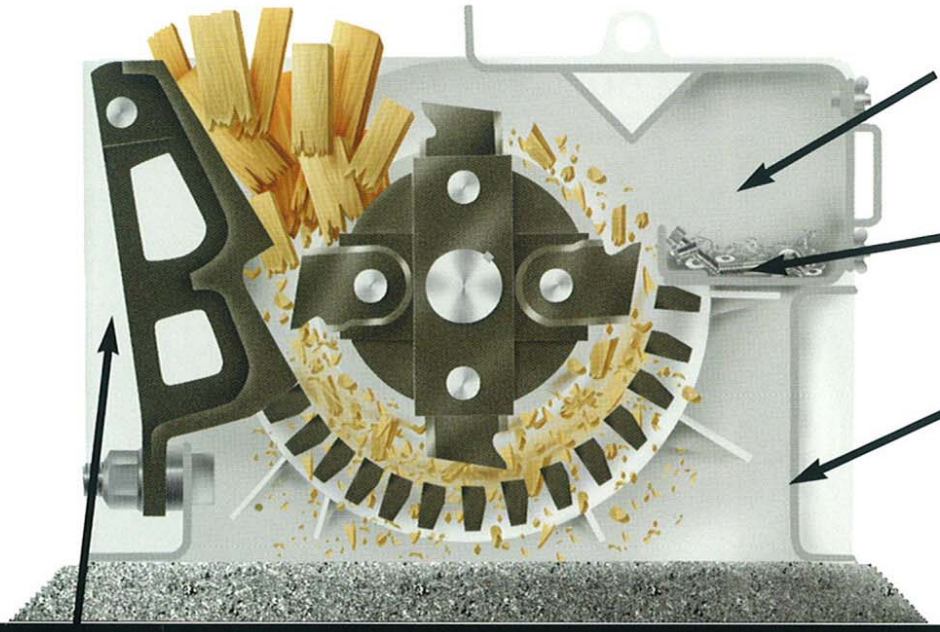


Second from the left is a heavy duty perforated screen plate. This is used in lieu of the bar section where very fine product is required.

Continuing from left to right is a cast manganese grid section, a large opening double bar grate, an individual bevel edged bar, and a narrow opening double bar. The grid section and double bar box grates are furnished for specialized applications.



THESE FEATURES MAKE WILLIAMS "NO-NIFE" HOGS... THE INDUSTRY STANDARD

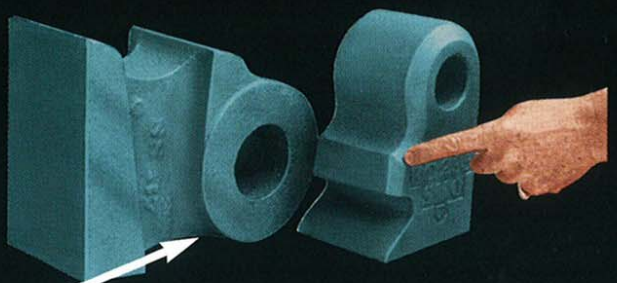


LINERS... Interior surfaces are protected with thick, easily replaceable liners to assure long service and to protect the frame.

TRAMP METAL TRAP... Standard on all "NO-NIFE" Hogs, the advanced design of William metal trap helps reduce damage due to uncrushables.

FRAME... Welded of steel. Superior rotor support and resistance to shock is provided by heavy rolled plate and welded structural shapes. Access section is completely removable without disturbing feed hopper or hood.

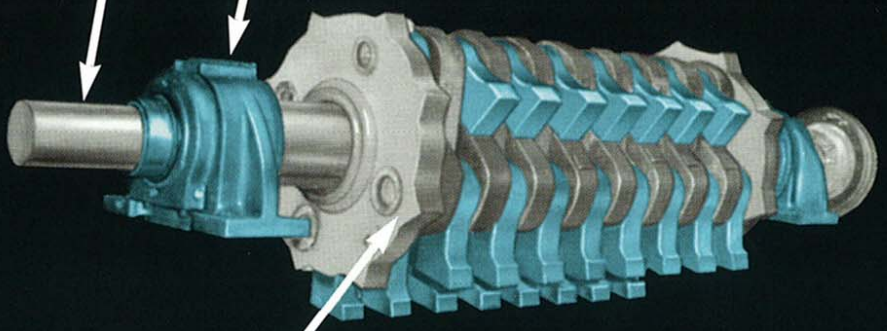
BREAKER PLATE... Massive manganese steel members. Externally adjustable design is available to optimize hogging action and compensate for wear.



HAMMERS... Massive heat treated SMO alloy hammers are furnished as standard.

SHAFT... Turned from alloy steel forgings, normalized and tempered. Shafts are oversized to reduce vibration.

BEARINGS... To insure long rotor life, heavy duty self aligning double row spherical roller bearing are furnished mounted in sealed housings.



HAMMER BOLTS... High tensile, heat treated, alloy steel, the bolts are surfaced hardened for resistance to wear.

FINAL RUNNING TEST... Each Hog is statically and dynamical balanced prior to shipment.

WILLIAMS WOOD HOGS/CONVEYORS



Large overrunning "No-Nife" Hog (1250 H.P. drive) used to process wood and bark into boiler fuel at 125 TPH.

Williams "No-Nife" Wood Hogs shred up to 300 TPH with these advantages:

- No Knives To Sharpen
- Excellent Particle Size Control
- Lower Power Requirement
- Internal Flywheel
- Easy Opening Cover
- Tramp Iron Pocket
- Oversized Shaft and Bearings
- High Surge Capacity

- **Sawmill • Boardmill • Pulp**
- **Paper • Millwork • Veneer**
- **Plywood • Lumber • Pallets**
- **Flooring • Dimension**
- **Particleboard • Hardboard**
- **Furniture • Cabinet • O.S.B.**
- **Construction/Demolition**

Williams 125 years of experience in manufacturing Hogs for the Forest Products Industry has resulted in a complete range of "No-Nife" Hog sizes and types to meet your individual requirements.

Williams' reputation for quality and reliability has resulted in the confidence of manufacturers in all aspects of wood handling and processing. A partial list of customers is shown on the back cover. Williams is truly the industry leader.

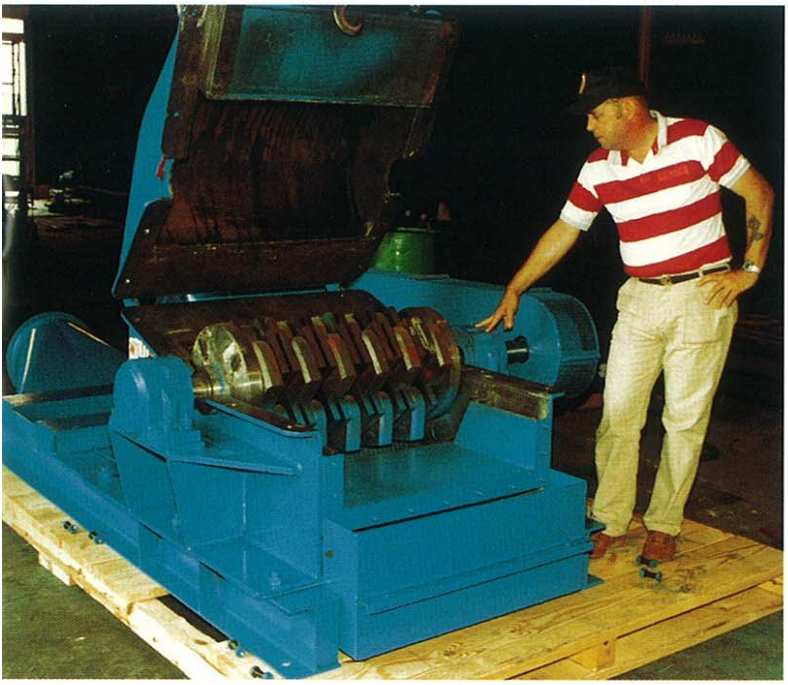
Rely on Williams know-how when planning your next installation.



Samples of shredded materials.

WILLIAMS HORIZONTAL FEED HOG

A Conveyor Fed, Horizontal Feed Hog For The Forest Products Industry



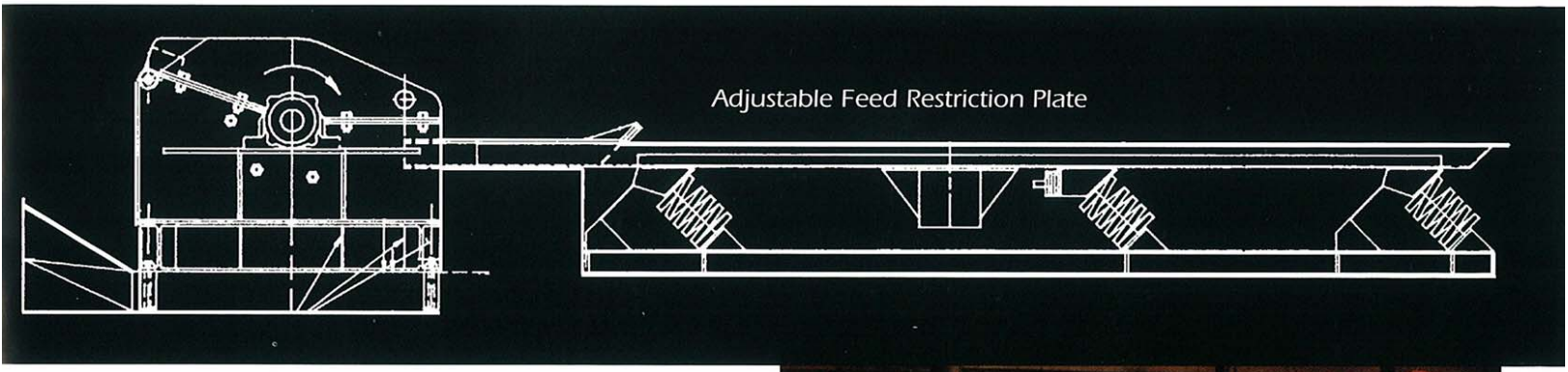


Complete And Easy Access

The hinged-top cover provides access to the entire machine interior for service and maintenance.

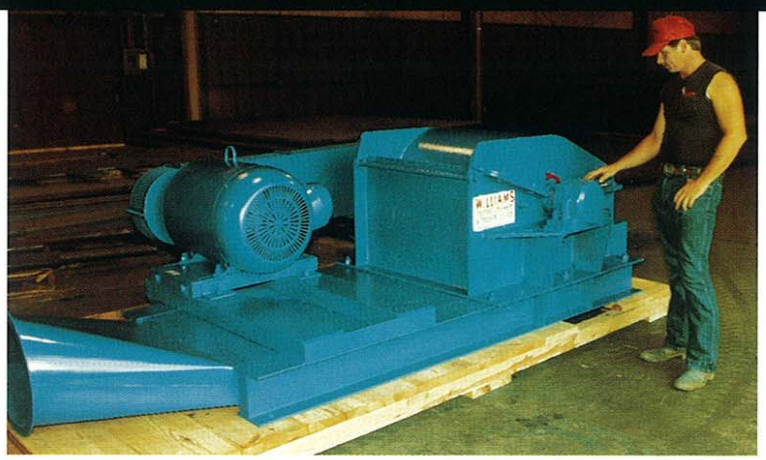
Self-Feed Hammers (No Pinch Roll Required)

Long edgings and other material are self-fed by the "No-Nife" Hog Hammers into the Grinding Chamber below the Horizontal Feed Line.



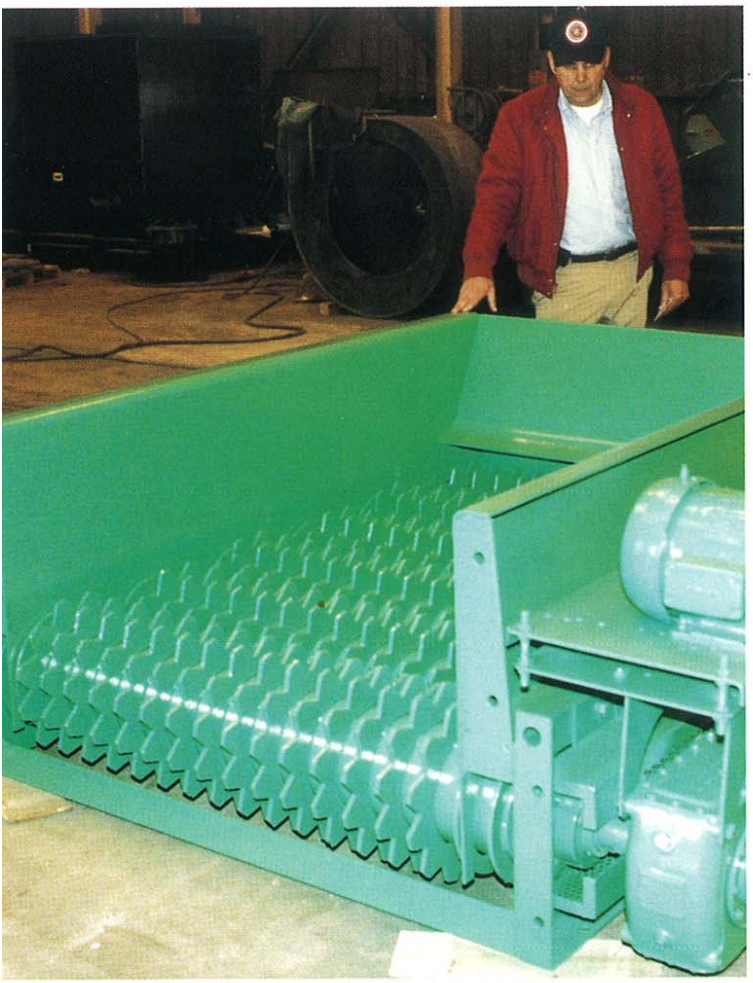
Heavy Duty Construction

- Steel Plate Housing and Structurals Welded Into Integral Units
- Oversized Alloy Steel Rotor Shaft
- Double Row Spherical Roller Bearings
- Swing Hammer of Cast Alloy Steel
- Cast Manganese Steel Grates (for longer wear life and greater strength)

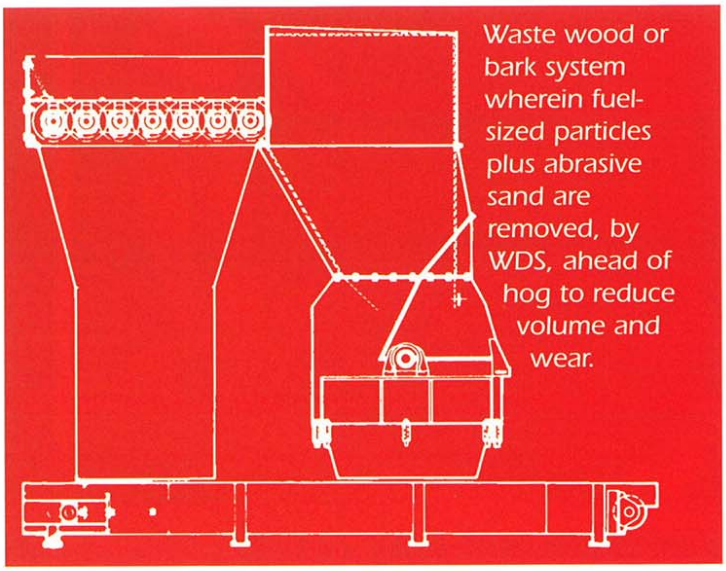


Air transport system recommended.

WILLIAMS ROTARY DISC SCREENS



Rotary Disc Screens are used for the consistent, high speed separation of oversized and undersized components of wood wastes. They Prequently are used as feeders into shredders or hogs, bypassing pre-sized materials.

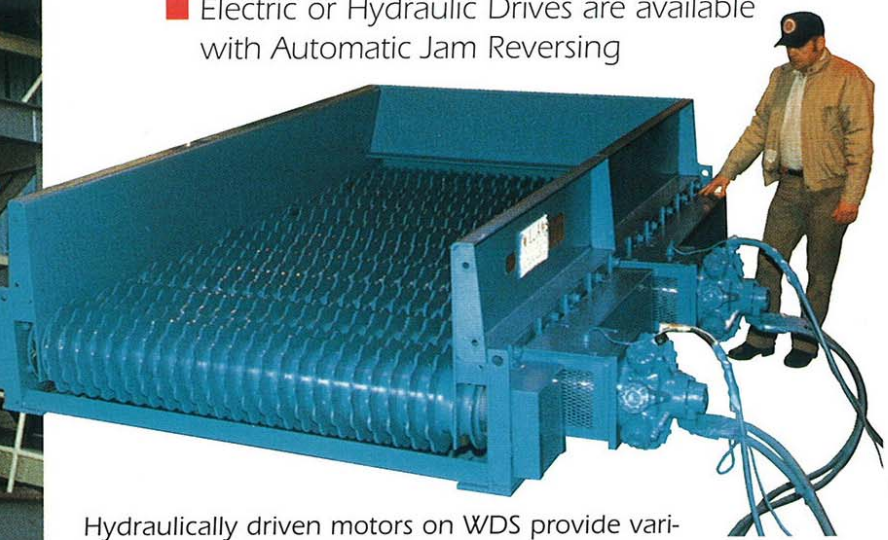


Williams Disc Screens (WDS®) offer these advantages:

- No Loosening of Shaft Assemblies
- No Laterial or Axial Shaft Movement
- No Dead Space at Sides of the Screen, Preventing Pile-up of Material
- Heavy Duty Construction
- Easy Maintenance
- Electric or Hydraulic Drives are available with Automatic Jam Reversing

Rotary disc screen feeding a Williams heavy duty wood hog at a large waste/energy facility.





Hydraulically driven motors on WDS provide variable speeds for more efficient screening and have rotational device which detects jams and reverses shafts to clear blockage.

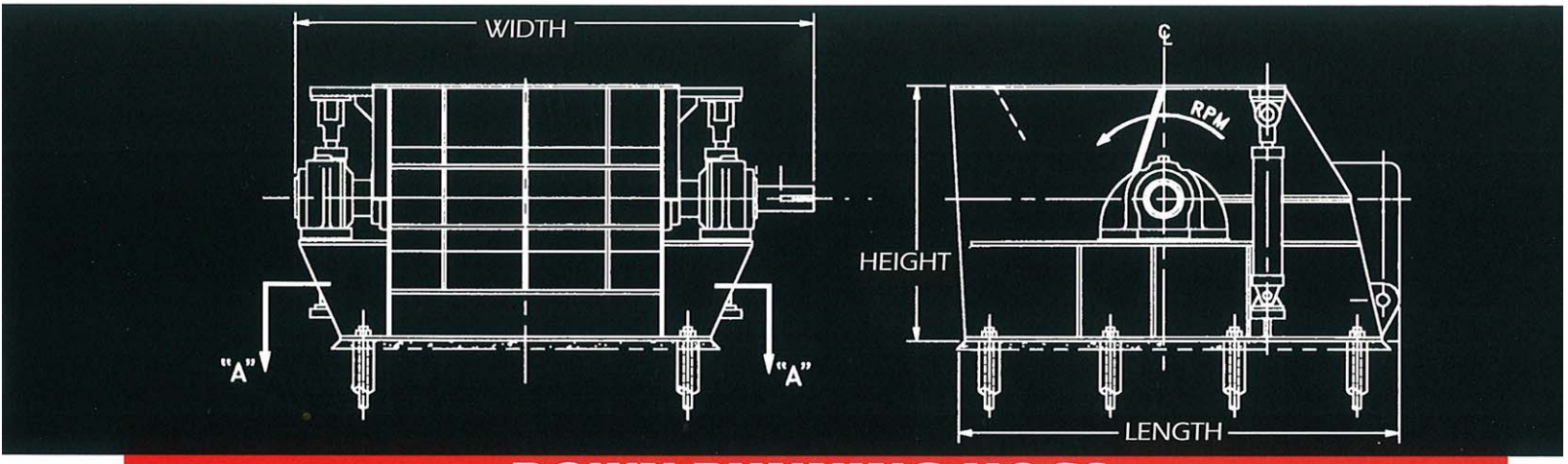
WILLIAMS PERFORMANCE & DIMENSIONS

Capacities and dimensions are intended as guides. Hog capacity will vary with type of wood or bark, moisture content, feed size, and uniformity of feed rate.

Capacities of 3" bark product are based on approximately 50% moisture content and include average amounts of broken pulpwood.

Capacities of minus 2" wood product apply to dry wood waste from manufacturing plants, green or dry sawmill refuse and damp or dry veneer and plywood waste.

Dimensions are subject to change. Use certified prints for final engineering layout.

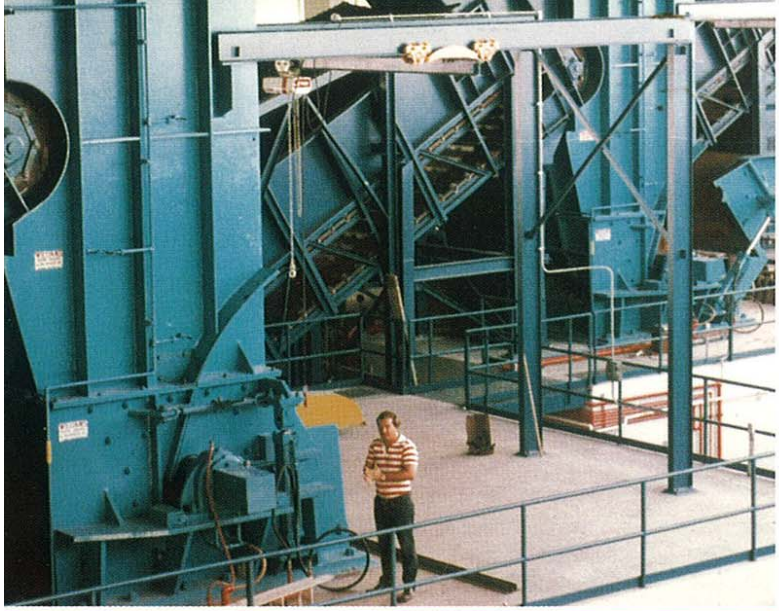


DOWN-RUNNING HOGS

SIZE	2 Inch Wood		3 Inch Bark		Usual RPM	Base Length (Inches)	Width (Inches)	Height (Inches)	Feed Opening (Inches)	Weight (Pounds)
	Capacity Tons/Hour	Usual Motor	Capacity Tons/Hour	Usual Motor						
C12	To 1	15-30	To 2	15-30	1800	50	51	31	12 x 19	2670
C17	1-2	25-40	4-6	25-40	1800	50	60	31	17 x 19	4140
C32	3-6	50-75	10-13	50-75	1800	50	77	31	32 x 19	5405
20 KS	3-6	50-75	10-13	50-75	1800	67	75	39	22 x 24	8165
30 KS	6-9	75-100	12-19	75-125	1800	67	87	39	32 x 24	9430
40 KS	8-12	100-150	18-26	100-150	1800	67	98	39	42 x 24	11270
60 KS	12-17	150-200	32-40	200-250	1800	67	120	39	62 x 24	16898
320	5-7	75-100	12-18	75-125	1800	78	76	45	22 x 29	11730
330	6-11	100-125	18-28	125-200	1800	78	88	45	32 x 29	13500
340	10-16	125-200	26-36	150-250	1800	78	100	45	42 x 29	15450
350	14-19	150-250	36-46	200-300	1800	78	112	45	52 x 29	17600
360	16-23	200-300	42-54	250-350	1800	78	122	45	62 x 29	20100
370	22-26	250-350	52-64	300-400	1200	78	132	45	72 x 29	23200
380	24-30	300-400	60-72	350-450	1200	78	147	45	82 x 29	26800
390	29-35	300-450	70-82	400-500	1200	78	157	45	92 x 29	31500
430	10-14	125-200	26-36	150-250	900	101	112	57	33 x 43	27025
445	14-23	200-250	36-54	250-350	900	101	127	57	48 x 43	32200
460	22-30	250-400	54-72	350-450	900	101	143	57	63 x 43	37950
475	29-38	350-500	72-90	400-600	900	101	154	57	78 x 43	45730
490	36-46	400-600	90-108	500-700	900	101	169	57	93 x 43	54050
540	22-26	250-400	48-62	300-400	900	109	124	66	46 x 65	44850
560	26-38	350-450	62-94	400-600	900	109	144	66	66 x 65	56240
580	38-52	450-700	96-125	600-800	900	109	164	66	86 x 65	70730
5100	52-65	600-800	120-156	700-1000	900	109	185	66	106 x 65	89130
5120	65-78	700-1000	150-187	800-1250	900	109	205	66	126 x 65	111550
640	24-32	300-400	60-78	350-500	720	128	137	86	46 x 69	63140
660	36-48	400-600	78-116	500-700	720	128	157	86	66 x 69	78660
680	48-64	600-800	114-156	700-1000	720	128	177	86	86 x 69	98900
6100	60-79	700-1000	156-194	900-1250	720	128	197	86	106 x 69	125350
6120	78-96	800-1250	192-234	1000-1500	720	128	217	86	126 x 69	158130
760	48-62	600-800	120-156	700-1000	720	153	170	116	65 x 80	143980
780	60-83	700-1000	156-206	900-1250	720	153	190	116	85 x 80	163530
7100	79-103	900-1250	204-258	1000-1500	720	153	210	116	105 x 80	189750
7120	96-125	1000-1500	240-312	1500-2000	720	153	230	116	125 x 80	218500

OVER-RUNNING HOGS

SIZE	1/2 Inch Wood		3/4 Inch Bark		Usual RPM	Base Length (Inches)	Width (Inches)	Height (Inches)	Feed Opening (Inches)	Weight (Pounds)
	Capacity Tons/Hour	Usual Motor	Capacity Tons/Hour	Usual Motor						
215 NF	To 2	15-30	To 5	15-30	1800	53	60	43	15 x 19	4650
220 NF	1-3	25-40	4-7	25-40	1800	53	67	43	20 x 19	4980
230 NF	2-4	50-75	6-10	50-75	1800	53	77	43	30 x 19	6000
240 NF	4-5	75-100	10-13	75-100	1800	53	87	43	40 x 19	7450
20 NF	2-4	50-75	7-10	50-75	1800	67	75	47	20 x 20	7100
30 NF	4-5	75-100	10-14	75-125	1800	67	87	47	30 x 20	8850
40 NF	5-7	100-150	14-19	125-200	1800	67	98	47	40 x 20	10800
50 NF	6-9	125-200	19-24	150-200	1800	67	108	47	50 x 20	12700
60 NF	8-11	150-200	24-29	200-250	1800	67	120	47	60 x 20	14650
20 GA	4-5	75-100	10-12	75-125	1200	69	76	54	20 x 40	10930
30 GA	5-7	100-125	14-18	125-200	1200	69	88	54	30 x 40	12650
40 GA	7-10	125-200	18-24	150-250	1200	69	100	54	40 x 40	14460
50 GA	8-12	150-250	24-30	200-300	1200	69	112	54	50 x 40	17020
60 GA	11-14	200-300	30-36	250-350	1200	69	122	54	60 x 40	19840
70 GA	12-17	250-350	36-42	300-400	1200	69	132	54	70 x 40	23000
80 GA	14-19	300-400	42-48	350-450	1200	69	147	54	80 x 40	27370
90 GA	17-22	300-450	48-54	400-500	1200	69	157	54	90 x 40	32000
430 GA	8-11	125-200	22-26	150-250	900	101	112	70	34 x 51	26700
445 GA	12-16	200-250	26-38	250-350	900	101	127	70	49 x 51	31500
460 GA	16-20	250-400	38-50	350-450	900	101	143	70	63 x 51	37100
475 GA	20-25	350-500	50-62	400-600	900	101	154	70	79 x 51	44700
490 GA	24-30	400-600	62-74	500-700	900	101	169	70	94 x 51	52900
540 GA	12-17	250-400	36-42	300-400	900	117	124	79	45 x 72	43800
560 GA	17-25	350-450	42-62	400-600	900	117	144	79	65 x 72	55000
580 GA	24-32	450-700	60-84	600-800	900	117	164	79	85 x 72	69200
5100 GA	32-41	600-800	84-102	700-1000	900	117	185	79	105 x 72	87100
5120 GA	41-49	700-1000	96-126	800-1250	900	117	205	79	125 x 72	109000
640 GA	16-20	300-400	42-52	350-500	720	136	124	90	46 x 69	61600
660 GA	20-30	400-600	48-78	500-700	720	136	144	90	66 x 69	77000
680 GA	30-40	600-800	72-102	700-1000	720	136	164	90	86 x 69	96600
6100 GA	40-50	700-1000	96-132	900-1250	720	136	184	90	106 x 69	122500
6120 GA	48-60	800-1250	120-156	1000-1500	720	136	204	90	126 x 69	154900



CUSTOMER SERVICE LABORATORY

WILLIAMS maintains a complete test facility and computer backed Engineering Department. We will make pilot runs on samples of your product and assist in the evaluation of Test Results.

Tests may be arranged by contacting our Representatives or by writing the factory.

Test Data is treated as customer's confidential information.



Apron pan, infeed conveyor with many years of service handling heavy waste.

With installations numbering in the thousands, WILLIAMS has practical experience to help you select and apply the proper wood or bark hog.

CONSIDER THIS PARTIAL LIST OF WILLIAMS USERS IN THE FOREST PRODUCTS INDUSTRY

American Can Corporation
Babcock & Wilcox Corporation
Bemis Manufacturing Company
Boise Cascade Corporation
Browning Ferris Corporation
Bowater Corporation
Canadian Forest Products Corporation
Champion Papers
Crown Zellerbach
Drexel
Georgia Pacific Corporation
International Paper Company
Kimberly Clark Corporation
Laidlaw & Company
Lazy Boy Furniture
MacMillan Bloedel, Inc.
Masonite Corporation
Olin Corporation
Potlatch Forest Group
Temple Inland, Inc.
Thomasville Furniture Industries
Union Camp Corporation
United Forest Products
Weyerhauser Corporation
Zurn/Nepco

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